Shourya Bafna

LinkedIn • Github • CodeChef • Codeforces • Leetcode • Personal Portfolio

PROFESSIONAL SUMMARY

Software engineering enthusiast with strong DSA expertise and hands-on experience building real-world applications. A 1200+ Codeforces rating and 300+ LeetCode solutions demonstrate strong problem-solving and analytical skills. Focused on designing scalable, reliable, and maintainable software while continuously strengthening engineering fundamentals, tackling complex challenges, and delivering practical solutions.

EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2024-2028	Bachelor of Technology (Computer Science and AI)	Newton School of Technology, Rishihood University	8.6/10.0
2023-2024	Intermediate (Class XII)	Satyameva Jayete International School	62.0%
2021-2022	Matriculation (Class X)	Sarala Birla Gyan Jyoti	92.0%

INTERNSHIPS

Full Stack Developer May 2025 - November 2025
RashtramAI *OnSite (Sonipat)*

Worked as a Full Stack Developer, a platform helping law students analyze constitutional topics and legal bills using RAG.

- Designed and developed end-to-end web features, integrating frontend interfaces with backend services for seamless user
- experience.
- Implemented RAG model integration to enable accurate retrieval of constitutional data and legal documents.
- Worked with React.js, Node.js, Express, MongoDB, PineconeDB, and APIs to deliver a high-performance and scalable user
- experience..
- Actively collaborating with the team to enhance performance, ensure data accuracy, and improve platform usability for law
- students.

Frontend Developer February 2025 - April 2025
Get Interview Confidence *Remote*

Interned as Frontend Developer, enhanced their website with refined design, faster performance, and engaging experience.

- Built and maintained web pages using HTML, CSS, JavaScript, and NextJS, ensuring cross-device compatibility.
- Collaborated with designers and backend developers to implement features and improve user workflows.
- Used Git/GitHub for version control, enabling smooth teamwork and code collaboration.

PROJECTS

Smart Waste Segregator (Github) October 2026

- **Tech Stack:** Python, Scikit-learn, Scikit-image, NumPy, Pandas, Gradio, SVM, Random Forest
- **Description:** Smart Waste Segregator is a classical ML-based image classifier that identifies and categorizes waste into six classes: cardboard, glass, metal, paper, plastic, and miscellaneous trash.
- Developed an image feature extraction pipeline using HSV color histograms, Histogram of Oriented Gradients (HOG), and Local Binary Patterns (LBP) to capture visual characteristics.
- Implemented and compared SVM and Random Forest classifiers using group-aware cross-validation, achieving **89.99% macro F1-score and 90.19% test accuracy** with the selected calibrated SVM model.

Next Flow (Github) (Demo) April 2026

- **Tech Stack:** TypeScript, Next.js, React Flow, Gemini API, Trigger.dev, PostgreSQL, Prisma
- **Description:** NextFlow is a powerful, visual node-based workflow automation builder. It allows users to create, connect, and execute complex workflows involving media processing and AI generation through an intuitive drag-and-drop interface.
- Built a visual workflow editor connecting LLM, text, image and video processing nodes.
- Integrated Gemini generation and asynchronous node execution with Trigger.dev; used FFmpeg and Sharp for media processing.

Vortex (Github) (Demo) June 2025

- **Tech Stack:** TypeScript, NodeJS, ExpressJS, Axios, Pinecone, OpenAI, node-cron
- **Description:** Vortex is an advanced AI-powered study assistant designed to help students learn more effectively. It
- combines a modern chat interface with powerful document processing capabilities to answer questions, solve math
- problems, and assist with study materials.
- **Features:** Automated Notes Retrieval, Vector Embeddings, Pinecone Integration, KaTeX Rendering, RAG Workflow

SKILLS

- **Languages:** Python, C++, TypeScript, JavaScript
- **ML & AI:** Machine Learning, Deep Learning (DL), RAG, Vector Embeddings, Quantization, Pandas, Scikit-learn, PyTorch, NumPy, TensorFlow
- **Databases:** Pinecone, MySQL, MongoDB, PostgreSQL, Prisma ORM
- **Developement:** React, Node.js, Express JS, HTML, CSS, Tailwind CSS, Django, Git and Github, Next JS